Abstract

Overall scientific questions.

Introduction

- Overview of the ISM. How dust forms, why we use CO as a tracer. Why we might use other tracers.
- Jeans collapse a model for star formation.
- Quasistatic turbulent core fragmentation as a model for star formation.
- IMF / CMF. How the high mass end changes. SF across different environments.
- (Nico and Alessio) Literature on decoupling cores from clumps. We would like to know at what stage the gas decouples.
- CHIMPS paper and other extragalactic attempts at this sort of work.

The interstellar medium (ISM) is made up of primarily gas, as well as dust and cosmic rays. The ISM can be divided up based on temperature, ionisation and density. It consists of 5 main phases: the cold molecular medium (CMM), which is defined such that the gas is molecular with a temperature range of $10-30\ K$ and a number density between $10^2-10^6\ cm^{-3}$; the cold neutral medium (CNM), which is defined such that the gas is mostly atomic with a temperature range of $30-100\ K$ and a number density between $1-100\ cm^{-3}$; the warm neutral medium (WNM), which is defined such that the gas is mostly atomic with a temperature range of $6000-10000\ K$ and number densities below $1\ cm^{-3}$; the warm ionised medium (WIM), which is defined such that the gas is mostly ionised with a temperature range of $8000-10000\ K$ and number densities around $0.4\ cm^{-3}$ and the hot ionised medium (HIM), which is defined such that the gas is mostly ionised with a temperature range of $10^6-10^7\ K$ and number densities below $10^{-2}\ cm^{-3}$ (see Klessen and Glover 2016). The focus of this work will be on the CMM.

Other than primordial elements, every atom formed inside a star. Therefore, initially, these elements that make up the gas are hot and have not formed in their molecular state. To get to the temperature of the CMM, the gas must be cooled. A star ejects matter throughout its life as stellar winds, as well as at the end of its life, in different ways depending on its mass. As the gas moves away from its progenitor, the number of ionising photons will decrease which will offset the balance between ionisation and recombination. At these higher temperatures, the cooling from Lyman-α becomes important. The general form of the cooling rate of a two-level system is shown below.

$$\Lambda_{ul} = \frac{g_u}{g_l} A_{ul} E_{ul} n_l e^{-E_{ul}/k_B T} \tag{1}$$

Equation 1 is the rearranged form of 2.38 in Tielens 2005. The formula comes about by assuming LTE and that the number density is much less than the critical density. g_u and g_l are statistical weights of the energy states, A_{ul} is the Einstein A coefficient of spontaneous emission, E_{ul} is the energy of the transition, and n_l is the number density of the lower state.

$$\Lambda_{HI} = 7.3 \times 10^{-19} e^{-118400/T} n_e n_H \tag{2}$$

Equation 2 appears as 2.70 in Tielens 2005. Above $10^4 K$, the exponential term stops being negligible, which causes the cooling rate to rise rapidly. This together with recombination to form atomic hydrogen is what facilitates the transition to the WNM. At higher metallicities the cooling is not dominated by Lyman-α but by other fine structure transitions from heavier elements. Once cooled, Equation 2 shows a ceiling for the temperature in the WNM above which the cooling rate can rapidly exceed the different heating rates (Tielens and Hollenbach 1985). A different fine structure transition dominates the cooling in the WNM and facilitates the transition to the CNM, the CII transition. The energy of this transition is $92K$ (Hollenbach and McKee 1989), meaning the exponent in Equation 1 becomes $e^{-92/T}$, allowing the gas to cool efficiently down to $\sim 100K$. The term in the exponent implies that below $100K$ the cooling from CII decreases rapidly. On the other hand, the first rovibrational transition ($J = 1$–0) of CO emits a photon with a rest frequency of $115.27GHz$ (Winnewisser et al. 1997), which gives an energy of $5.53K$, using $h\nu/k_B$. This allows the cooling rate of CO to dominate in this temperature regime and allow the gas to get to a temperature of around $10K$. Even though CO is a more efficient coolant, CII seems to be a more dominant cooling mechanism at low metallicity as well as for diffuse gas with densities less than $10^3 cm^{-3}$ (Glover and Clark 2012). This is partly because at low metallicity, the CO cannot be shielded by the dust and photodissociates by way of the ISRF. It is also important to note that Glover and Clark 2012 indicate that if the cloud is shielded from the effects of photoelectric heating then cooling from CII can allow the gas to get down to temperatures of $20K$ without the presence of CO.

Once there is gas at the typical temperatures of the CMM, molecular hydrogen can begin to form. Molecular hydrogen should form directly between two atoms of hydrogen but the Einstein A coefficient is too small for the process to allow for a photon to be emitted during their collision (Wakelam et al. 2017). This means that molecular hydrogen cannot form efficiently this way, therefore another mechanism is required to get the amount of molecular hydrogen expected. The formation process involves atomic hydrogen sticking to a dust grain and colliding with another stuck atom of hydrogen allowing for the recombination of molecular hydrogen. The energy produced by this is enough to break the bonds between the dust grain and the molecule of hydrogen, allowing for it to return to the CMM (Wakelam et al. 2017). This has been confirmed experimentally using various dust grain substitutes made of carbon or silicates. Katz et al. 1999 find that the peak efficiency of the recombination occurs below $20K$ for both types of grain. The formation of CO is very complex (see Glover et al. 2010) and can involve different pathways, the simplest of which is the formation from a carbon atom and a hydroxyl radical. Even though it is thought that the gas being in a molecular state is not necessary for star formation, the cloud being able to shield itself from the ISRF is paramount to the formation of both stars and molecular gas (Glover and Clark 2012). They find that a cloud of atomic gas is still capable of forming stars at the same rate as molecular gas is able to form. This however does not change the fact that molecular hydrogen still traces star formation, it just may not trace all star formation.

Tracing molecular gas is important because it allows for the understanding of the steps that lead to star formation. CO is regarded as a good tracer of molecular gas for its abundance in molecular clouds and spontaneous low energy transitions, like the one described above. The idea is to trace the more abundant molecular hydrogen, which is not directly possible due to its lack of emission at the typical temperatures of the CMM. The lowest possible rotational transition of molecular hydrogen is at an energy of $510K$ (Wakelam et al. 2017). CO has been the gold standard when it comes to tracing molecular hydrogen, but tracers like HCN or N_2H^+ have been found to trace the dense gas much more effectively than CO(Kauffmann et al. 2017). N_2H^+ is almost always being detected above column densities of $10^{22} cm^{-2}$ (Fehér et al. 2025), meaning it is much better at tracing the clumps where stars are likely to be forming.

Regions of the ISM that contain the CMM are also known as molecular clouds. These clouds are the densest constituents of the ISM, the collapse of which leads to star formation. For a cloud to collapse the gravitational force pulling the gas together must overcome that of the internal pressure of the gas. Hydrostatic equilibrium is the point at which both forces cancel one another out. This allows for the derivation of the mass above which a spherical system would collapse, known as the Jeans mass.

$$M_J = \frac{\pi^{3/2} {c_s}^3}{{\rho_o}^{1/2} G^{3/2}} \tag{3}$$

Equation 3 appears as the rearranged form of 9.23 in Stahler and Palla 2004 using the fact that $M_J \sim \rho_o \lambda^3$. c_s refers to the isothermal sound speed in the medium given by $\sqrt{k_B T/\mu m_p}$. This equation assumes that the system is isolated with no external forces, but the ISM is turbulent therefore an external pressure could be present. In this instance the external pressure would oppose the internal pressure of the gas, making it easier for the gas to collapse. This mass is known as the Bonner-Ebert mass.

$$M_{BE} = \frac{1.18 {c_s}^4}{{P_o}^{1/2} G^{3/2}} \tag{4}$$

Equation 4 appears as 9.16 in Stahler and Palla 2004. P_o is the external pressure on the cloud. There are several mechanisms that affect star formation that these equations do not account for. Magnetic fields oppose the gravitational collapse of clouds, slowing or stopping the collapse of clouds (Krasnopolsky and Gammie 2005). This allows more gas to be accreted before collapse leading to higher mass star formation. Stellar feedback strips the diffuse gas from the cloud slowing their growth but allowing the dense centres to continue to be star forming (Watkins et al. 2025). It can also cause the destruction of the host cloud. Watkins et al. 2025 also found that stellar feedback likely breaks apart larger clouds into smaller clouds. Protostellar radiative feedback stops fragment formation causing most of the gas to be accreted onto a few objects leading to higher mass star formation (Krumholz, Klein, and McKee 2007). Cloud-cloud collisions between both atomic and molecular clouds facilitates the formation of massive molecular cloud cores, making them the dominant mechanism for forming high mass stars. Additionally, there is observational evidence that they are responsible for starbursts (e.g. the Antenna galaxies) (Fukui et al. 2021).

As a cloud collapses, the Jeans mass scales proportionally with temperature and density $\left(M_J \propto \rho^{-1/2}T^{3/2}\right)$, as shown by Equation 3. During gravitational collapse the density always increases and providing the gas is isothermal, the Jeans mass decreases as the density increases (Ward-Thompson and Whitworth 2011). Therefore, sections of the collapsing cloud become Jeans unstable and begin gravitational collapse causing multiple collapsing regions. These collapsing regions can go on to fragment again. Eventually the density increases so much so heat is unable to radiate away and cooling becomes unable to keep the gas isothermal, and so temperature increases. Assuming the collapsing cloud acts like a polytrope, it will follow the equation of state $(P = K\rho^{\gamma})$, where γ is the adiabatic index (Chieze 1987). If heat cannot radiate away then the collapse can be considered adiabatic, so the temperature is now a function of density $\left(T = K\rho^{\gamma-1}\right)$. This now means that the Jeans mass only scales proportionally with density $\left(M_J \propto \rho^{(3/2)(\gamma-4/3)}\right)$. In molecular clouds, the gas is primarily diatomic or monatomic which have 5 and 3 degrees of freedom, which gives them adiabatic indices of 5/3 and 7/5, respectively $(\gamma = 1 + 2/d.o.f.)$. Both these indices result in a direct proportionally between the Jeans mass and density, causing fragmentation to cease.

The above picture of fragmentation is not complete as it does not take into account turbulence or magnetic fields. Magnetic fields as described above stop fragmentation by exerting a force on charged particles outwards. The charged particles then collide with neutral matter causing their velocities to couple (Krasnopolsky and Gammie 2005). It is for this reason that ambipolar diffusion was introduced, which allows the star forming material to be decoupled from the ions affected by strong magnetic fields. This occurs when the clouds get sufficiently dense that cosmic rays and the ISRF are unable to cause substantial ionisation, which reduces the rate at which the collisions occur (Mestel and Spitzer 1956). This allows the charged particles to pass through the neutral matter without imposing its velocity, allowing the neutral matter to go on to fragment and form stars.

Ambipolar diffusion is a slow process and is unable to reproduce the numerical formation time of a core from observations ($\sim 1Myr$) (Ciolek and Basu 2001). This can be solved with the addition of supersonic turbulence (Mac Low and Klessen 2004). The dominant driving process of this turbulence in star forming galaxies is supernovae, which self regulates by temporarily suppressing star formation, leading to high mass star formation and thus more supernovae. These high mass stars during their lifetimes produce ionising radiation that contributes short lasting turbulence by heating the gas which then contracts as it cools (Kritsuk and Norman 2002). Gravitational instabilities are a local driving mechanism of the turbulence but are unable to delay collapse for much more than a free-fall time due to the fact that the turbulence decays. Magnetorotational instabilities allow for turbulent flows to be generated from shear by magnetic fields generating Maxwell stress, which may occur at the outskirts of the Milky Way disc. The supersonic turbulence generated by these driving forces builds up gas at its shock front, forming dense sheets and filaments. These dense regions even form in the presence of magnetic fields. These driving forces produce random motions in the gas that can act against gravitational collapse temporarily slowing the infall. This supersonic turbulence can only do so as long as it is continuously driven because it decays quickly. For an isothermal shock the density scales with the Mach number $\left(\rho' = \mathcal{M}^2\rho\right)$, where the Mach number is directly proportional to the rms velocity dispersion due to turbulence. If the turbulence is treated as an additional pressure then the effective sound speed can be defined which takes into account the rms velocity dispersion $(c^2_{\text{s,eff}} = c^2_s + \nu^2_{\text{rms}}/3)$ (Chandrasekhar 1951). It is assumed that $\nu_{\text{rms}} >> c_s$ because the turbulence is supersonic, which means that the Jeans mass now is a function of the turbulence as well as the density $\left(M_J \propto \nu_{\text{rms}}^2 \rho^{-1/2}\right)$. This means that an increase in turbulence increases the Jeans mass, preventing collapse. Eventually the turbulence decays and the cloud can collapse. If the clump is able to become sufficiently dense that it decouples from the ambient gas in the cloud, a second shock will be unable to disrupt it. If the gas is not able to collapse in this time then a second shock would disrupt the clump inducing further turbulence and preventing collapse further. These successive shocks in a short amount of time is the basis for high mass star formation in this theory (see Mac Low and Klessen 2004).

IMF

Project Work

- Motivations (link back to the lit review).
- Methods.
- Conclusions and what that might mean.
- Future work.

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